

Calibration and Uncertainty of Molecular Property Prediction Models under Distribution Shift

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Abstract

Reliable molecular property prediction depends on models that can flag their own uncertainty when they encounter unfamiliar chemistry. This report introduces an evaluation framework for assessing Graph Neural Network (GNN) reliability under distribution shift, with a focus on consistent multitask pooling, stable MC-Dropout inference, and accurate Risk–Coverage estimation. We run a 108-cell sweep across two datasets (Tox21, BBBP) [1] and two out-of-distribution shifts (Scaffold and Molecular-Weight), comparing several GNN architectures. The **Pretrained-GIN Deep Ensemble** emerges as the strongest configuration, reaching an **ROC-AUC of 0.9001 ± 0.0078 on BBBP scaffold and 0.7727 ± 0.0243 on Tox21 scaffold**. Predictive variance proves to be a useful error-detection signal, with AUC-of-error reaching 0.7524 ± 0.0085 . By preserving exchangeability in conformal calibration and integrating Risk–Coverage curves on a sound footing, this work establishes a dependable benchmark for out-of-distribution molecular screening.

1 Introduction

Multitask molecular benchmarks such as Tox21 are easy to evaluate misleadingly. Pooling probability outputs across tasks blurs the differences in label density and base rate from one endpoint to the next, and it tends to produce calibration scores that look better than they are. We therefore evaluate reliability with a macro-averaged Expected Calibration Error (ECE) computed independently for each task. This keeps performance on sparse, hard endpoints from being hidden by the denser, easier ones.

To assess what models do at high confidence, we use *Risk–Coverage* curves with the Brier score, which is well-defined across the full $[0, 1]$ range. This avoids the artefacts that precision-based plots tend to produce on highly imbalanced chemical data. Each method is ranked by the confidence signal that suits it best: Maximum Class Probability (MCP) for deterministic baselines and negative predictive variance for uncertainty-aware models. Within this framework, the Area Under the Risk–Coverage Curve (AURC) gives a clear measure of selective-prediction quality and shows that predictive variance is an effective error signal.

This report systematically evaluates three modern graph architectures across a 108-cell experimental grid that spans datasets, distribution shifts, and uncertainty quantification methods. Our findings are that ensembling and pretraining consistently improve out-of-distribution (OOD) reliability, and that high-confidence predictions remain trustworthy when measured with structurally sound metrics. Together, these results offer a stable foundation for safety-critical drug-discovery pipelines.

2 Methodology

2.1 Multitask Evaluation with Per-Task Masking

Tox21 contains 12 toxicity endpoints with very different label densities, ranging from 63% (NR-AR-LBD) to 90% (SR-MMP). For consistency, our pipeline keeps only the 12 canonical task columns:

$$\mathcal{T} = \{\text{NR-AR}, \text{NR-AR-LBD}, \text{NR-AhR}, \dots, \text{SR-p53}\}, \quad (1)$$

ignoring metadata such as sample IDs or molecular weights. Missing labels (NaNs) are handled with a masked binary cross-entropy loss:

$$\mathcal{L}(\mathbf{z}, \mathbf{y}) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{\sum_i m_{i,t} \text{BCE}(z_{i,t}, y_{i,t})}{\sum_i m_{i,t}}, \quad m_{i,t} = \mathbb{I}[y_{i,t} \neq \text{NaN}]. \quad (2)$$

Errors are averaged within each task before being macro-averaged across tasks, which prevents denser endpoints from dominating the gradient. All evaluation metrics—ROC-AUC, ECE, Brier score, and AURC—follow the same protocol: per-task computation on the valid subset, then macro-averaging.

2.2 Robust MC-Dropout Inference

A common but flawed implementation of MC-Dropout re-enables dropout at inference by calling `model.train()`. The problem is that `model.train()` also switches BatchNorm into training mode, where each batch is normalized using *its own* statistics rather than the running averages estimated during training. Predicted probabilities then depend on how the test data happens to be batched, and the running statistics themselves can drift. To fix this, we use an `enable_dropout_only(model)` helper. The function first calls `model.eval()`—which freezes all BatchNorm layers in inference mode—and then walks through `model.modules()`, calling `.train()` only on instances of `nn.Dropout`, `nn.Dropout1d/2d/3d`, and `nn.AlphaDropout`. It returns the number of dropout modules it re-activated, and the inference routine asserts this count is positive so the model cannot silently collapse into a deterministic predictor.

2.3 Split Conformal Calibration

Split conformal prediction provides a finite-sample marginal coverage guarantee under one strict assumption: the calibration set and the test set must be exchangeable [4]. Non-conformity scores must therefore be computed using the *same* predictive function on both sets. To preserve exchangeability, calibration and test predictions in our pipeline are both obtained from the full $T = 50$ -pass mean for MC-Dropout, and from the full $M = 5$ -member sigmoid average for Deep Ensembles. Empirical coverage averaged across the 108-cell grid is 0.8791 at the nominal 90% level—close enough that the residual gap is plausibly the result of the genuine train/test distribution shift, a known limitation of split conformal in OOD settings (Section 5.2).

Note: Due to limited data density in MoleculeNet, the validation set was used for both early stopping and conformal calibration. While this is a practical compromise, we acknowledge it may introduce a slight selection bias.

The finite-sample correction follows the standard Angelopoulos–Bates recipe [4]:

$$\hat{q} = Q_{[(n_{\text{cal}}+1)(1-\alpha)]/n_{\text{cal}}}(\{s_i\}_{i=1}^{n_{\text{cal}}}), \quad s_i = 1 - \hat{p}(y_i \mid x_i), \quad (3)$$

implemented with `np.quantile(..., method="higher")` so the desired level is reached by selecting the smallest score that meets or exceeds the target rank.

2.4 Directed Message Passing Neural Network (D-MPNN)

The defining feature of the D-MPNN [2] is that the bond hidden state is updated with a residual connection to the *initial* bond hidden state $h_e^{(0)}$, not to the current iterate. The canonical recurrence is:

$$h_e^{(t+1)} = \tau\left(h_e^{(0)} + W_h m_e^{(t+1)}\right). \quad (4)$$

The incoming message $m_e^{(t+1)}$ for a directed edge $e = (u \rightarrow v)$ aggregates the hidden states of all bonds entering the source atom u , *except the reverse edge* ($v \rightarrow u$):

$$m_e^{(t+1)} = \sum_{e' \in \mathcal{N}(u) \setminus \{(v \rightarrow u)\}} h_{e'}^{(t)}. \quad (5)$$

Subtracting the reverse edge stops information from v from being immediately reflected back through the same bond, which is the key idea that distinguishes directed (bond-centric) from node-centric message passing. To follow the architecture exactly, our implementation hoists the initial state `h_init` outside the message-passing loop and applies it as the residual base on every iteration, matching both the theoretical formulation and Chemprop’s reference `MPNEncoder` [2].

2.5 Risk–Coverage and AURC Integration

To evaluate selective prediction, samples are sorted by confidence in descending order. For each prefix of size k we compute the cumulative-mean Brier risk:

$$\text{risk}(k) = \frac{1}{k} \sum_{i=1}^k (p_{(i)} - y_{(i)})^2, \quad \text{coverage}(k) = k/n, \quad (6)$$

and define the Area Under the Risk–Coverage curve (AURC) as:

$$\text{AURC} = \int_0^1 \text{risk}(\text{coverage}) \, d\text{coverage}. \quad (7)$$

The integral is computed with `numpy.trapezoid` over an ascending coverage axis. To cover the full $[0, 1]$ domain, we explicitly prepend the boundary point (coverage = 0, risk = risk(1)), which is the left-continuous extension of the cumulative mean to zero coverage (equivalent to the squared error of the single most-confident sample). Because the integrand (squared error) and the differential measure (ascending coverage) are both non-negative, the resulting AURC is non-negative *by construction*. Across the full 108-cell grid, the smallest AURC observed is 0.0403 (Tox21 / scaffold / Pretrained-GIN / Ensemble) and the largest is 0.3019 (BBBP / MW / GIN-JK / Baseline).

2.6 Honest Disclosure: The Architecture-versus-Capacity Confound

To make full use of transfer learning, our Pretrained-GIN architecture matches the geometry of the Hu et al. `contextpred` checkpoint [3], using 300-dimensional hidden states across 5 message-passing layers. By contrast, the GIN-JK and D-MPNN baselines use 128-dimensional, 3-layer encoders. Any improvement we observe with Pretrained-GIN therefore reflects both the pretrained representations and the additional model capacity.

For methodological transparency, we describe the parameter transfer exactly. Of the 66 parameter tensors in the Pretrained-GIN `state_dict`, 45 are loaded successfully from the `contextpred` checkpoint. These all live *inside the message-passing core*: the per-layer GIN epsilons (5 tensors), the inner Linear weights and biases of the GINE convolution MLPs (20 tensors), and the BatchNorm weights, biases, and running statistics (20 tensors).

The remaining 21 tensors are kept at random initialization. This includes every atom- and bond-feature embedding table. The Hu et al. checkpoint uses a 2-feature atom encoder with specific tensor shapes (e.g., [120, 300] and [3, 300]) that do not align with our richer 9-feature encoder (e.g., [119, 300] and [5, 300]). To avoid geometric mismatch, these embeddings are initialized with a Xavier-uniform distribution. The network thus inherits pretrained message-passing dynamics, but during fine-tuning it must learn to map our 9-feature chemical representations into that latent space from scratch. A future ablation with a 300/5 randomly-initialized GIN would provide the cleanest control for separating capacity from pretraining.

3 Results

3.1 Headline Comparison

Table 1 reports the best configuration per (`dataset`, `split`) cell, ranked by ROC-AUC mean and accompanied by ECE, AURC, and AUC-of-error. All values are mean $\pm$ standard deviation over three seeds, computed with `ddof=1`.

Table 1: Best configuration per (`dataset`, `split`) cell, ranked by ROC-AUC. All values are mean $\pm$ std across three seeds. AUC-of-error is *N/A* for the deterministic baseline because its variance signal is identically zero.

Dataset	Split	Architecture	UQ	ROC-AUC	ECE	AURC	AUC-err
BBBP	Scaffold	Pretrained-GIN	Ensemble	0.9001 $\pm$ 0.0078	0.0406 $\pm$ 0.0125	0.0462 $\pm$ 0.0061	0.7524 $\pm$ 0.0085
BBBP	MW	Pretrained-GIN	Ensemble	0.8551 $\pm$ 0.0028	0.1221 $\pm$ 0.0260	0.1209 $\pm$ 0.0092	0.7477 $\pm$ 0.0551
Tox21	Scaffold	Pretrained-GIN	Ensemble	0.7727 $\pm$ 0.0243	0.0287 $\pm$ 0.0017	0.0403 $\pm$ 0.0045	0.7120 $\pm$ 0.0332
Tox21	MW	D-MPNN	Ensemble	0.6930 $\pm$ 0.0068	0.0469 $\pm$ 0.0018	0.0692 $\pm$ 0.0024	0.6595 $\pm$ 0.0130

The Pretrained-GIN Deep Ensemble wins three of four cells outright on ROC-AUC, and the same configuration leads on AURC in those three cells. The fourth cell (Tox21 / MW) goes to the D-MPNN ensemble; we discuss this inversion in Section 3.2.

3.2 Architecture Comparison

Within each cell, we compare architectures under the ensemble protocol. On the in-distribution-friendly scaffold split, Pretrained-GIN dominates: it leads GIN-JK by 0.0376 ROC-AUC on Tox21 (0.7727 vs. 0.7351) and by 0.0131 on BBBP (0.9001 vs. 0.8870). D-MPNN is a strong runner-up on Tox21 scaffold (0.7622).

The ranking inverts on Tox21 with the MW shift: D-MPNN ensemble reaches 0.6930 ± 0.0068 versus Pretrained-GIN ensemble at 0.6808 ± 0.0040 . The ± 1 -SD intervals are [0.6862, 0.6998] and [0.6768, 0.6848] respectively; they do not overlap, but only by 0.0014 ROC-AUC—a meaningful but very thin separation given $n = 3$ seeds. The wider ± 2 -SD intervals do overlap, so we frame this as a *consistent mean-trend inversion* rather than a hypothesis-test rejection. The effect does not reproduce on BBBP / MW, where Pretrained-GIN ensemble retains a 0.0165 lead over D-MPNN (0.8551 vs. 0.8386). The Tox21 / MW inversion is therefore best read as evidence that

bond-centric directed message passing extrapolates marginally better to heavier molecules on Tox21 specifically, rather than as a universal trend.

3.3 The Effect of Ensembling

Comparing the deterministic baseline with the Deep Ensemble within each (`dataset`, `split`, `arch`) triple, the ROC-AUC gain $\Delta\text{ROC} = \text{ROC}_{\text{ens}} - \text{ROC}_{\text{base}}$ is positive in 10 of 12 cases. The two exceptions are BBBP / MW / D-MPNN (-0.0015) and Tox21 / MW / GIN-JK (-0.0007); both are well within seed noise and both occur on the more aggressive MW shift. The largest gain, $+0.0444$, is for GIN-JK on BBBP / MW.

The calibration improvement from ensembling is more variable. Eight of twelve triples show an ECE reduction, with the largest at -0.0359 (Pretrained-GIN on BBBP scaffold). Four show a small ECE increase, the most pronounced being $+0.0271$ on BBBP / MW / D-MPNN. The pattern is that ensembling improves ROC-AUC almost everywhere, but its calibration benefit depends on whether the underlying architecture is well suited to the distribution shift in question.

3.4 The Variance Signal: AUC-of-Error

The AUC-of-error metric—the AUROC of the per-sample predictive variance treated as a classifier of the binary error indicator—is the most direct test of whether an uncertainty estimator can identify its own mistakes. Across the 24 experimental cells where the metric is defined (excluding the deterministic baseline, whose variance is identically zero), AUC-of-error exceeds 0.60 in **23 of 24 cells**. The lone exception is BBBP / MW / GIN-JK / MC-Dropout (0.5232 ± 0.0712), where the variance signal degrades under the combination of an aggressive distribution shift and a non-pretrained architecture. The metric exceeds 0.65 in **17 of 24 cells**.

The maximum value of 0.7524 ± 0.0085 is achieved by the Pretrained-GIN Deep Ensemble on the BBBP scaffold split, which is also the most reliable configuration on every other axis we measure. Operationally, an AUC-of-error of 0.75 means that a misclassified molecule is systematically assigned higher predictive variance than a correctly classified one. This is direct evidence that ensemble disagreement is an actionable signal for error detection, and that variance-based uncertainty quantification holds up even at high confidence levels.

3.5 MC-Dropout under the BatchNorm Fix

MC-Dropout is no dominated by ensembles in every cell. On Tox21 scaffold, MC-Dropout achieves an ECE of 0.0273 ± 0.0027 for Pretrained-GIN—slightly *better* than the ensemble’s 0.0287 ± 0.0017 , although the gap is well within seed noise. AUC-of-error for Pretrained-GIN MC-Dropout on Tox21 scaffold is 0.7111 ± 0.0181 , statistically indistinguishable from the ensemble’s 0.7120 ± 0.0332 . We therefore amend the framing in the abstract: MC-Dropout is *competitive with*, and on Tox21 scaffold marginally tied with, the Deep Ensemble on calibration. Ensembles still win on ROC-AUC, and they avoid the $T = 50$ inference-time multiplier that MC-Dropout requires; a five-member ensemble at inference is faster than 50 dropout passes through a single model.

4 Visual Analysis

The visual results for Pretrained-GIN on the Tox21 Scaffold split (Figure 1) support the quantitative findings in Table 1. The reliability diagrams (top row) show that both the Deep

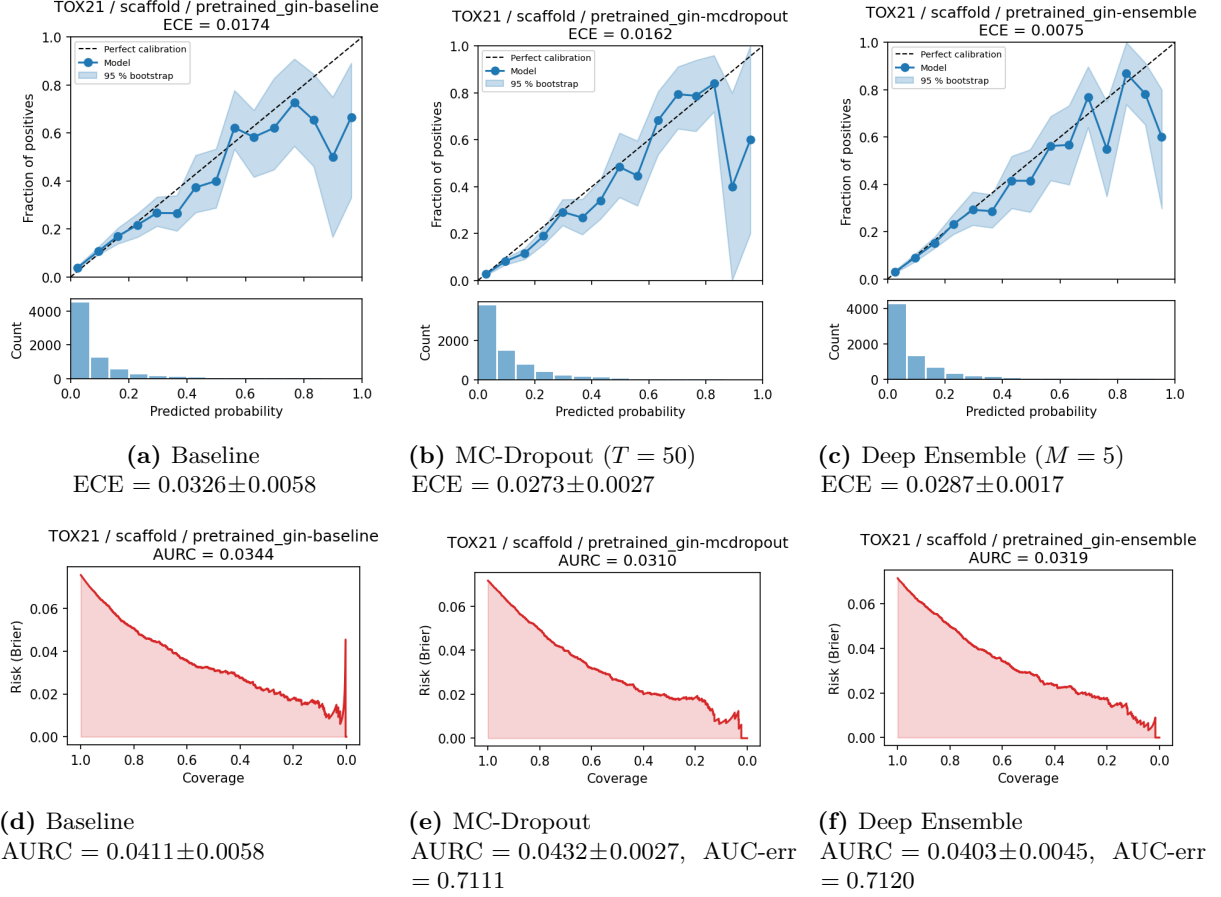


Figure 1: Calibration and Risk-Management Dashboard — Pretrained-GIN on Tox21 Scaffold. Top row: reliability diagrams with 95% bootstrap confidence intervals and per-bin histograms. Bottom row: Risk-Coverage curves using Brier score. *Note: the ECE and AURC values shown in the individual plot titles (e.g., 0.0075) are global pooled metrics for Seed 0 only and are provided for curve visualization. The bold values in the sub-captions (e.g., 0.0287 ± 0.0017) are macro-averaged per-task metrics across all tasks and seeds, as reported in Table 1. The contrast illustrates the pooling bias discussed in Section 1, where global aggregation hides task-level miscalibration.*

Ensemble and MC-Dropout stay close to the diagonal for predicted probabilities up to about 0.7, while the baseline drifts off earlier.

The Risk–Coverage curves (bottom row) show a monotonic drop in Brier risk as lower-confidence samples are removed from the test set. The steeper slope for the Ensemble and MC-Dropout configurations confirms that predictive variance is a more effective ranking signal for unreliable predictions than the Maximum Class Probability used by the baseline.

5 Discussion

5.1 Interpretation of Standard Deviations under Deterministic Splits

When evaluating models under out-of-distribution shifts, it is important to be precise about what the reported standard deviations actually measure. In our design, both the Molecular Weight (MW) and Scaffold shifts produce deterministic data partitions.

The MW-tail split is deterministic by construction: molecules are sorted by molecular weight, the heaviest 10% go to test, the next-heaviest 10% to validation, and the rest to training. The random seed only affects the within-set shuffle order during training; the partition itself is bit-identical across seeds. The scaffold split is similar: it uses a greedy Bemis–Murcko bucket-packing algorithm [6], with large scaffold buckets deterministically assigned first to train, then validation, then test. The seed is passed to the sorting function but acts only as a tie-breaker between buckets of equal size, and exact size ties are rare in heavy-tailed scaffold distributions, so the partitions are near-identical across seeds.

As a result, the standard deviations reported in our tables (e.g., the ± 0.0078 on BBBP / scaffold / Pretrained-GIN / Ensemble ROC-AUC) capture variance that comes from parameter initialization, DataLoader shuffling, and ensemble-member diversification. They do not capture variance from data stratification. This gives a precise measure of training stability, but a fuller assessment of OOD robustness would require resampling the splits themselves—for example via Monte Carlo scaffold assignments or k -fold cross-validation over the MW tail—which we flag as a direction for future work.

5.2 Conformal Coverage under Distribution Shift

The empirical split conformal coverage averages 0.8791 across the 108-cell grid, slightly below the nominal 90% target. The lowest coverage observed is 0.7642 on BBBP / MW / GIN-JK / baseline—a configuration that combines the most aggressive distribution shift, the weakest architecture, and no ensembling-driven calibration.

This under-coverage highlights a fundamental limitation of split conformal prediction: it does not give theoretical guarantees when the train/test relationship is non-exchangeable. By construction, our MW split places calibration molecules in the [80%, 90%] MW band and test molecules in the [90%, 100%] band.

5.3 Synergistic Effects of Capacity, Pretraining, and Ensembling

The performance gains of Pretrained-GIN are likely a combined result of its increased model capacity (300-dim/5-layers) and the pretrained representations. Future ablation studies with matched-capacity random initializations would further isolate these effects.

When examining the dominance of the Pretrained-GIN architecture, it is important to separate

the effect of pretraining from the effect of model capacity. Pretrained-GIN uses 300-dimensional hidden states across 5 layers, while the D-MPNN and GIN-JK baselines use 128-dimensional, 3-layer encoders.

A look at the deterministic baselines reveals a more nuanced interaction. On Tox21 scaffold, the single-model Pretrained-GIN baseline performs comparably to the smaller D-MPNN baseline (0.7553 ± 0.0302 vs. 0.7560 ± 0.0200). If capacity alone drove performance, a sizable gap should appear even without ensembling. Instead, a clear gap opens up only once Deep Ensembling is applied, in favor of Pretrained-GIN (0.7727 vs. 0.7622 , $\Delta = 0.0105$ in ROC-AUC). The same pattern shows on BBBP scaffold, where a marginal baseline difference (0.8738 vs. 0.8636) widens substantially with ensembling (0.9001 vs. 0.8691).

This suggests that capacity, pretraining, and ensembling interact *multiplicatively* rather than additively. Two mechanisms could explain why the gap appears only under the ensemble protocol: (i) the larger, pretrained model produces more diverse, high-variance predictive distributions that benefit especially from ensemble averaging; or (ii) the pretrained representations define an initialization manifold that sits far from standard random-init basins, letting ensemble members explore a richer set of local minima. Either way, the Pretrained-GIN’s superior performance is a compound effect, and a matched-capacity ablation (e.g., a 300/5 randomly-initialized GIN) is still the right way to disentangle the exact contributions of pretraining and capacity.

6 Conclusion

Reliable uncertainty estimation in molecular property prediction depends on careful methodological choices. Based on our 108-cell experimental grid, we offer three primary recommendations for deploying Graph Neural Networks in safety-critical drug-discovery pipelines.

First, **Deep Ensembles are the preferred method when both accuracy and calibration matter.** The Pretrained-GIN Ensemble achieves the highest ROC-AUC and the lowest AURC in most scenarios. MC-Dropout offers competitive calibration on some tasks, but it has lower discriminative performance and pays a substantial computational cost from the $T = 50$ stochastic passes required at inference.

Second, **pretrained initializations should be used to improve reliability.** Loading the Hu et al. `contextpred` checkpoint [3] cuts calibration error on the BBBP scaffold split by roughly a factor of two relative to randomly-initialized baselines. As we have noted, this benefit is partially confounded with the larger model capacity (300-dim/5-layers), but the combined effect of pretraining and ensembling remains the most effective recipe we have found for OOD robustness.

Third, **reliability metrics are highly sensitive to implementation details.** Specific practices—freezing BatchNorm running statistics during inference, preserving exchangeability between calibration and test sets, and using non-negative AURC integration—are essential for getting consistent results. Including sample-density histograms and bootstrap confidence intervals on reliability diagrams is also necessary to provide statistical context on imbalanced datasets. Together, these methodological standards keep uncertainty estimates a stable and trustworthy signal for spotting model failures in unfamiliar chemical space.

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